POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5032 | Semester V | Theory | 4 Credits

Power Electronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5032 as Power Electronics in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Delhi’s Power Electronics course and polytechnic power electronics curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Converter designs, triggering circuit parameters, inverter PWM settings and UPS/SMPS configurations must be based on approved manufacturer data, certified equipment testing, current safety standards and competent professional review.

Verified source note: Verified sources support power devices (SCR, Triac, GTO, BJT, MOSFET, IGBT), controlled rectifiers, AC voltage controllers, cycloconverters, DC-DC converters (buck, boost, Cuk) and inverters. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Power Electronics studies the application of solid-state electronics for the control and conversion of electric power. It develops the ability to analyze power semiconductor devices (SCR, MOSFET, IGBT), evaluate phase-controlled rectifiers, design DC-DC converters (choppers) and inverters, and understand frequency conversion techniques while respecting the technical and safety boundaries of high-power electronic systems.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Basic electronics, electrical circuit theory, analog electronics, electrical machines and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the construction, characteristics and applications of power semiconductor devices like SCR, MOSFET and IGBT.
2. Explain the working and performance of single-phase and three-phase controlled rectifiers with various loads.
3. Explain the principles of DC-DC converters (choppers) and frequency conversion using cycloconverters and AC voltage controllers.
4. Explain the operation and PWM techniques of single-phase and three-phase inverters for industrial and domestic applications.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the characteristics, triggering and commutation of power semiconductor devices.	Understand
CO2	Explain the performance of phase-controlled rectifiers and AC voltage controllers.	Understand
CO3	Explain the working principles and control strategies of DC-DC converters (choppers).	Understand
CO4	Explain the operation of inverters and their applications in motor drives and UPS.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Power Semiconductor Devices and Triggering	Power diodes; Thyristors (SCR, Triac, GTO); Power BJT, MOSFET and IGBT; VI characteristics; two-transistor model of SCR; triggering methods (R, RC, UJT); commutation techniques (natural and forced); series and parallel operation.	Approx. 16 hours	CO1	Module 1
2	Phase Controlled Rectifiers and AC Controllers	Single-phase half-wave and full-wave controlled rectifiers (R, RL loads); effect of freewheeling diode; three-phase controlled rectifiers; single-phase and three-phase AC voltage controllers; cycloconverters (single-phase).	Approx. 16 hours	CO2	Module 2
3	DC-DC Converters (Choppers)	Principles of chopper operation; step-up and step-down choppers; control strategies (TRC and CLC); chopper classifications (Class A to E); non-isolated converters (Buck, Boost, Buck-Boost); Cuk converter.	Approx. 14 hours	CO3	Module 3
4	Inverters and Applications	Single-phase and three-phase voltage source inverters (VSI); current source inverters (CSI); 180 and 120 degree conduction modes; Pulse Width Modulation (PWM) techniques (single, multiple, sinusoidal); UPS and SMPS.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Power Semiconductor Devices and Triggering

Power diodes; Thyristors (SCR, Triac, GTO); Power BJT, MOSFET and IGBT; VI characteristics; two-transistor model of SCR; triggering methods (R, RC, UJT); commutation techniques (natural and forced); series and parallel operation.

CO1 · Approx. 16 hours

Phase Controlled Rectifiers and AC Controllers

Single-phase half-wave and full-wave controlled rectifiers (R, RL loads); effect of freewheeling diode; three-phase controlled rectifiers; single-phase and three-phase AC voltage controllers; cycloconverters (single-phase).

CO2 · Approx. 16 hours

DC-DC Converters (Choppers)

Principles of chopper operation; step-up and step-down choppers; control strategies (TRC and CLC); chopper classifications (Class A to E); non-isolated converters (Buck, Boost, Buck-Boost); Cuk converter.

CO3 · Approx. 14 hours

Inverters and Applications

Single-phase and three-phase voltage source inverters (VSI); current source inverters (CSI); 180 and 120 degree conduction modes; Pulse Width Modulation (PWM) techniques (single, multiple, sinusoidal); UPS and SMPS.

CO4 · Approx. 14 hours

MODULE 1

Power Semiconductor Devices and Triggering

Power diodes; Thyristors (SCR, Triac, GTO); Power BJT, MOSFET and IGBT; VI characteristics; two-transistor model of SCR; triggering methods (R, RC, UJT); commutation techniques (natural and forced); series and parallel operation.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

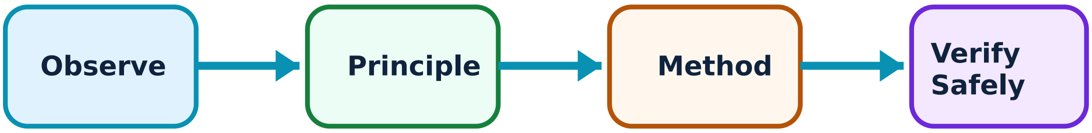
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power Semiconductor Devices and Triggering: identify the system, state its principle, follow the correct method and verify the result safely.

1. SCR and power switches−

The Silicon Controlled Rectifier (SCR) is the foundational power switch. MOSFETs and IGBTs provide faster switching for high-frequency applications. Understanding their VI characteristics, turn-on (triggering) and turn-off (commutation) mechanisms is essential for designing efficient power converters.

Study and application method

Sketch the VI characteristics of an SCR and label the holding and latching currents; compare the switching speeds of a Power MOSFET and an IGBT.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the VI characteristics of an SCR and label the holding and latching currents; compare the switching speeds of a Power MOSFET and an IGBT.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Power devices dissipate significant heat. Do not operate devices without authorised heat sinks and thermal protection as per manufacturer ratings.

2. Triggering and commutation−

Triggering involves applying a gate pulse to turn on the SCR. Commutation is the process of turning it off by reducing the current below the holding level. Natural commutation occurs in AC circuits, while forced commutation (using LC circuits) is required for DC circuits.

Study and application method

Describe the UJT pulse triggering circuit for an SCR; compare Class A and Class B forced commutation techniques conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the UJT pulse triggering circuit for an SCR; compare Class A and Class B forced commutation techniques conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Triggering circuits must be isolated from the high-power circuit (e.g., using pulse transformers or opto-isolators) to prevent damage to control electronics.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Phase Controlled Rectifiers and AC Controllers

Single-phase half-wave and full-wave controlled rectifiers (R, RL loads); effect of freewheeling diode; three-phase controlled rectifiers; single-phase and three-phase AC voltage controllers; cycloconverters (single-phase).

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

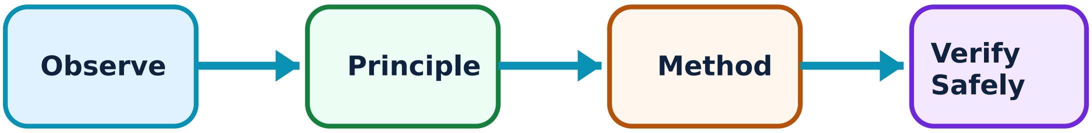
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Phase Controlled Rectifiers and AC Controllers: identify the system, state its principle, follow the correct method and verify the result safely.

1. Controlled rectifiers−

Phase-controlled rectifiers convert AC to variable DC by controlling the firing angle (alpha). Single-phase full converters provide two-quadrant operation. Three-phase rectifiers are used for high-power industrial applications like DC motor drives.

Study and application method

Sketch the output voltage waveform of a single-phase full-bridge controlled rectifier for a resistive load; explain the role of a 'freewheeling diode' in RL loads.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the output voltage waveform of a single-phase full-bridge controlled rectifier for a resistive load; explain the role of a 'freewheeling diode' in RL loads.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Controlled rectifiers produce harmonics and poor power factor at high firing angles. Use authorised filters and compensation techniques for industrial installations.

2. AC voltage controllers and cycloconverters−

AC voltage controllers vary the RMS voltage of an AC supply at the same frequency. Cycloconverters convert AC at one frequency to AC at another (usually lower) frequency without a DC link, used in low-speed, high-torque drives.

Study and application method

Describe the working of a single-phase AC voltage controller using antiparallel SCRs; list two applications of a single-phase to single-phase cycloconverter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of a single-phase AC voltage controller using antiparallel SCRs; list two applications of a single-phase to single-phase cycloconverter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: AC-AC conversion involves complex switching patterns. Do not implement these circuits without authorised control logic and snubber circuits for protection.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

DC-DC Converters (Choppers)

Principles of chopper operation; step-up and step-down choppers; control strategies (TRC and CLC); chopper classifications (Class A to E); non-isolated converters (Buck, Boost, Buck-Boost); Cuk converter.

Approx. 14 hours **CO3** **Understand · Apply**

Learning goals

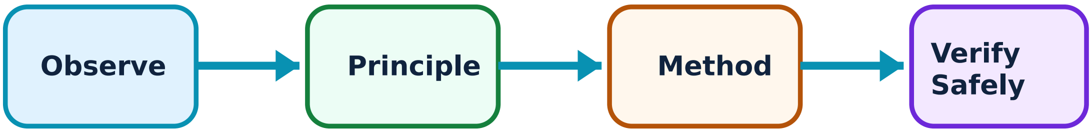
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DC-DC Converters (Choppers): identify the system, state its principle, follow the correct method and verify the result safely.

1. Chopper principles and control−

Choppers convert fixed DC to variable DC by high-speed switching. Time Ratio Control (TRC) varies the duty cycle (on-time/period). Classifications (Class A-E) define the quadrants of operation in the voltage-current plane.

Study and application method

Explain the principle of a step-down chopper; compare pulse-width modulation (PWM) and frequency modulation control strategies.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of a step-down chopper; compare pulse-width modulation (PWM) and frequency modulation control strategies.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Choppers involve high-frequency switching and EMI (Electromagnetic Interference). Use authorised shielding and layout techniques to minimize noise.

2. Buck, Boost and Buck-Boost converters−

Buck converters step down voltage, Boost converters step up voltage, and Buck-Boost can do both. These are the building blocks of modern power supplies (SMPS) and battery management systems, providing high efficiency and compact size.

Study and application method

Sketch the circuit diagram of a Boost converter and derive the output voltage expression conceptually; list the advantages of a Cuk converter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the circuit diagram of a Boost converter and derive the output voltage expression conceptually; list the advantages of a Cuk converter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inductive energy storage in choppers can cause high voltage spikes. Ensure all converters have authorised protection diodes and snubber circuits.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Inverters and Applications

Single-phase and three-phase voltage source inverters (VSI); current source inverters (CSI); 180 and 120 degree conduction modes; Pulse Width Modulation (PWM) techniques (single, multiple, sinusoidal); UPS and SMPS.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

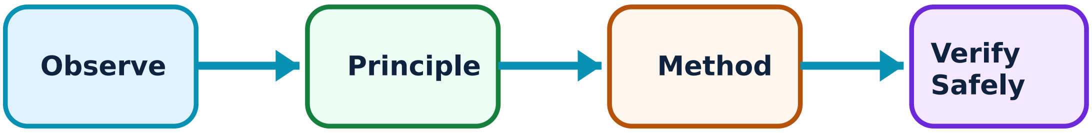
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Inverters and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. Inverter operation and PWM−

Inverters convert DC to AC. Voltage Source Inverters (VSI) maintain a constant DC voltage. PWM techniques control the output voltage and frequency while reducing harmonic content. Sinusoidal PWM (SPWM) is widely used for motor drives and renewable energy integration.

Study and application method

Compare the 180-degree and 120-degree conduction modes of a three-phase VSI; describe the principle of Sinusoidal PWM (SPWM).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the 180-degree and 120-degree conduction modes of a three-phase VSI; describe the principle of Sinusoidal PWM (SPWM).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Inverters can produce high dV/dt which stresses motor insulation. Use authorised filters or cable lengths as per drive-motor specifications.

2. Power electronics applications−

Applications include Uninterruptible Power Supplies (UPS) for critical loads, Switched Mode Power Supplies (SMPS) for electronics, and variable speed drives for motors. These systems improve energy efficiency and provide precise control of electrical energy.

Study and application method

Sketch a block diagram of an online UPS system; identify the role of power electronics in a grid-connected solar PV system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a block diagram of an online UPS system; identify the role of power electronics in a grid-connected solar PV system.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

> **Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

> **Common mistake / precaution:** UPS and SMPS systems involve stored energy in capacitors and batteries. Follow authorised safety procedures for maintenance and disposal of power electronic assemblies.

> **Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

> State symbols and SI units before substitution.

Recall 2

> Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Power Semiconductor Devices and Triggering

Use the concept flow to revise observation, principle, method and verification.

Module 2: Phase Controlled Rectifiers and AC Controllers

Use the concept flow to revise observation, principle, method and verification.

Module 3: DC-DC Converters (Choppers)

Use the concept flow to revise observation, principle, method and verification.

Module 4: Inverters and Applications

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare three power semiconductor devices in a conceptual table showing their ratings and applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the output waveforms for a single-phase half-controlled rectifier with RL load.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the output voltage conceptually for a Buck converter given the input voltage and duty cycle.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the conduction sequence for a three-phase inverter in 180-degree mode.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the power electronic components in a domestic inverter or computer SMPS.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Power Semiconductor Devices and Triggering

Explain SCR and power switches and state one correct method, application or precaution.−

The Silicon Controlled Rectifier (SCR) is the foundational power switch. MOSFETs and IGBTs provide faster switching for high-frequency applications. Understanding their VI characteristics, turn-on (triggering) and turn-off (commutation) mechanisms is essential for designing efficient power converters.

Answer framework: Sketch the VI characteristics of an SCR and label the holding and latching currents; compare the switching speeds of a Power MOSFET and an IGBT.

Important point: Power devices dissipate significant heat. Do not operate devices without authorised heat sinks and thermal protection as per manufacturer ratings.

Explain Triggering and commutation and state one correct method, application or precaution.−

Triggering involves applying a gate pulse to turn on the SCR. Commutation is the process of turning it off by reducing the current below the holding level. Natural commutation occurs in AC circuits, while forced commutation (using LC circuits) is required for DC circuits.

Answer framework: Describe the UJT pulse triggering circuit for an SCR; compare Class A and Class B forced commutation techniques conceptually.

Important point: Triggering circuits must be isolated from the high-power circuit (e.g., using pulse transformers or opto-isolators) to prevent damage to control electronics.

Module 2 — Phase Controlled Rectifiers and AC Controllers

Explain Controlled rectifiers and state one correct method, application or precaution. −

Phase-controlled rectifiers convert AC to variable DC by controlling the firing angle (alpha). Single-phase full converters provide two-quadrant operation. Three-phase rectifiers are used for high-power industrial applications like DC motor drives.

Answer framework: Sketch the output voltage waveform of a single-phase full-bridge controlled rectifier for a resistive load; explain the role of a 'freewheeling diode' in RL loads.

Important point: Controlled rectifiers produce harmonics and poor power factor at high firing angles. Use authorised filters and compensation techniques for industrial installations.

Explain AC voltage controllers and cycloconverters and state one correct method, application or precaution.−

AC voltage controllers vary the RMS voltage of an AC supply at the same frequency. Cycloconverters convert AC at one frequency to AC at another (usually lower) frequency without a DC link, used in low-speed, high-torque drives.

Answer framework: Describe the working of a single-phase AC voltage controller using antiparallel SCRs; list two applications of a single-phase to single-phase cycloconverter.

Important point: AC-AC conversion involves complex switching patterns. Do not implement these circuits without authorised control logic and snubber circuits for protection.

Module 3 — DC-DC Converters (Choppers)

Explain Chopper principles and control and state one correct method, application or precaution.−

Choppers convert fixed DC to variable DC by high-speed switching. Time Ratio Control (TRC) varies the duty cycle (on-time/period). Classifications (Class A-E) define the quadrants of operation in the voltage-current plane.

Answer framework: Explain the principle of a step-down chopper; compare pulse-width modulation (PWM) and frequency modulation control strategies.

Important point: Choppers involve high-frequency switching and EMI (Electromagnetic Interference). Use authorised shielding and layout techniques to minimize noise.

Explain Buck, Boost and Buck-Boost converters and state one correct method, application or precaution.−

Buck converters step down voltage, Boost converters step up voltage, and Buck-Boost can do both. These are the building blocks of modern power supplies (SMPS) and battery management systems, providing high efficiency and compact size.

Answer framework: Sketch the circuit diagram of a Boost converter and derive the output voltage expression conceptually; list the advantages of a Cuk converter.

Important point: Inductive energy storage in choppers can cause high voltage spikes. Ensure all converters have authorised protection diodes and snubber circuits.

Module 4 — Inverters and Applications

Explain Inverter operation and PWM and state one correct method, application or precaution.−

Inverters convert DC to AC. Voltage Source Inverters (VSI) maintain a constant DC voltage. PWM techniques control the output voltage and frequency while reducing harmonic content. Sinusoidal PWM (SPWM) is widely used for motor drives and renewable energy integration.

Answer framework: Compare the 180-degree and 120-degree conduction modes of a three-phase VSI; describe the principle of Sinusoidal PWM (SPWM).

Important point: Inverters can produce high dV/dt which stresses motor insulation. Use authorised filters or cable lengths as per drive-motor specifications.

Explain Power electronics applications and state one correct method, application or precaution.−

Applications include Uninterruptible Power Supplies (UPS) for critical loads, Switched Mode Power Supplies (SMPS) for electronics, and variable speed drives for motors. These systems improve energy efficiency and provide precise control of electrical energy.

Answer framework: Sketch a block diagram of an online UPS system; identify the role of power electronics in a grid-connected solar PV system.

Important point: UPS and SMPS systems involve stored energy in capacitors and batteries. Follow authorised safety procedures for maintenance and disposal of power electronic assemblies.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Power Semiconductor Devices and Triggering.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Phase Controlled Rectifiers and AC Controllers.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: DC-DC Converters (Choppers).
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Inverters and Applications.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Power diodes; Thyristors (SCR, Triac, GTO); Power BJT, MOSFET and IGBT; VI characteristics; two-transistor model of SCR; triggering methods (R, RC, UJT); commutation techniques (natural and forced); series and parallel operation..
2. Develop a complete answer or practical plan for: Single-phase half-wave and full-wave controlled rectifiers (R, RL loads); effect of freewheeling diode; three-phase controlled rectifiers; single-phase and three-phase AC voltage controllers; cycloconverters (single-phase)..
3. Develop a complete answer or practical plan for: Principles of chopper operation; step-up and step-down choppers; control strategies (TRC and CLC); chopper classifications (Class A to E); non-isolated converters (Buck, Boost, Buck-Boost); Cuk converter..
4. Develop a complete answer or practical plan for: Single-phase and three-phase voltage source inverters (VSI); current source inverters (CSI); 180 and 120 degree conduction modes; Pulse Width Modulation (PWM) techniques (single, multiple, sinusoidal); UPS and SMPS..

LAST-MILE REVISION

Quick Revision

Module 1: Power Semiconductor Devices and Triggering

Power diodes; Thyristors (SCR, Triac, GTO); Power BJT, MOSFET and IGBT; VI characteristics; two-transistor model of SCR; triggering methods (R, RC, UJT); commutation techniques (natural and forced); series and parallel operation.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Phase Controlled Rectifiers and AC Controllers

Single-phase half-wave and full-wave controlled rectifiers (R, RL loads); effect of freewheeling diode; three-phase controlled rectifiers; single-phase and three-phase AC voltage controllers; cycloconverters (single-phase).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: DC-DC Converters (Choppers)

Principles of chopper operation; step-up and step-down choppers; control strategies (TRC and CLC); chopper classifications (Class A to E); non-isolated converters (Buck, Boost, Buck-Boost); Cuk converter.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Inverters and Applications

Single-phase and three-phase voltage source inverters (VSI); current source inverters (CSI); 180 and 120 degree conduction modes; Pulse Width Modulation (PWM) techniques (single, multiple, sinusoidal); UPS and SMPS.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
SCR and power switches	The Silicon Controlled Rectifier (SCR) is the foundational power switch.
Triggering and commutation	Triggering involves applying a gate pulse to turn on the SCR.
Controlled rectifiers	Phase-controlled rectifiers convert AC to variable DC by controlling the firing angle (alpha).
AC voltage controllers and cycloconverters	AC voltage controllers vary the RMS voltage of an AC supply at the same frequency.
Chopper principles and control	Choppers convert fixed DC to variable DC by high-speed switching.
Buck, Boost and Buck-Boost converters	Buck converters step down voltage, Boost converters step up voltage, and Buck-Boost can do both.
Inverter operation and PWM	Inverters convert DC to AC.
Power electronics applications	Applications include Uninterruptible Power Supplies (UPS) for critical loads, Switched Mode Power Supplies (SMPS) for electronics, and variable speed drives for motors.

LEARNING RESOURCES

References

Recommended power-electronics references

1. M. H. Rashid, Power Electronics: Circuits, Devices and Applications.
2. P. S. Bimbhra, Power Electronics.
3. M. D. Singh and K. B. Khanchandani, Power Electronics.
4. Ned Mohan, T. M. Undeland and W. P. Robbins, Power Electronics: Converters, Applications and Design.

Approved public power-electronics sources

- https://nptel.ac.in/courses/108102145
- https://onlinecourses.nptel.ac.in/noc20_ee97/preview

These sources provide the verified study basis while official Course 5032 detail is unavailable; they do not replace official equipment manuals, authorised power-converter testing, certified designs or authorised industrial protocols.